1D CNN — Architecture Variant Comparison

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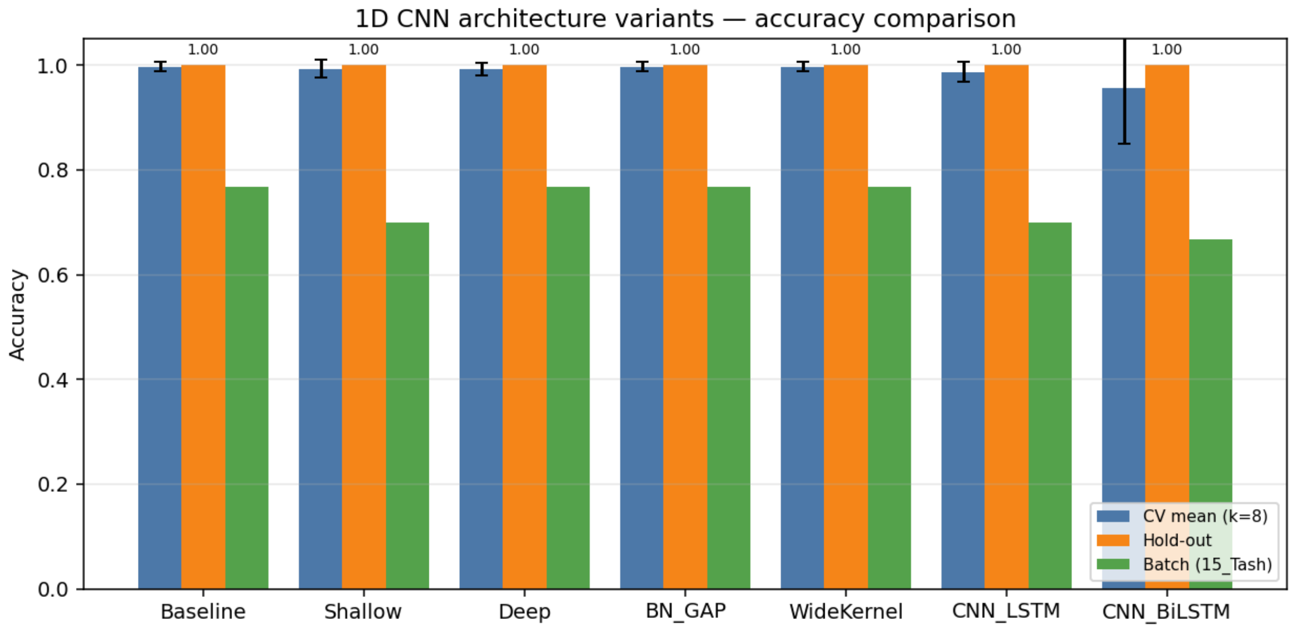
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	8 · True
Augmentation	on · copies=2 · amplitude_scale, baseline_offset
Epochs · batch size	60 · 16

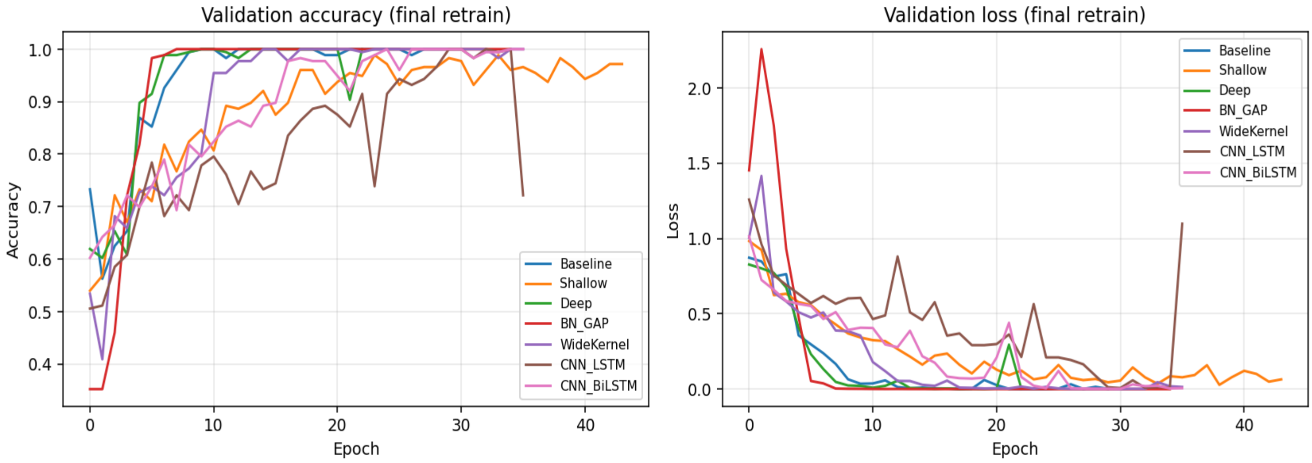
Variant comparison (summary)

Variant	Params	CV mean (k=8)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,190	0.9966	0.0089	1.0000	0.0021	0.7667	23/30
Shallow	72,454	0.9932	0.0179	1.0000	0.0095	0.7000	21/30
Deep	203,878	0.9931	0.0119	1.0000	0.0005	0.7667	23/30
BN_GAP	57,190	0.9966	0.0089	1.0000	0.0001	0.7667	23/30
WideKernel	131,622	0.9966	0.0089	1.0000	0.0008	0.7667	23/30
CNN_LSTM	68,390	0.9865	0.0191	1.0000	0.0091	0.7000	21/30
CNN_BiLSTM	105,510	0.9560	0.1063	1.0000	0.0011	0.6667	20/30

Best on hold-out: **Baseline** · Best on CV: **Baseline**



Training curves (final retrain on full training pool)



Validation accuracy (final retrain)
Accuracy
Epoch
Baseline
Shallow
Deep
BN_GAP
WideKernel
CNN_LSTM
CNN_BiLSTM
Validation loss (final retrain)
Loss
Epoch

Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,190
CV mean ± std	0.9966 ± 0.0089
Hold-out test acc / loss	1.0000 · 0.0021
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	765	37	1.0000	1.0000	1.0000
2	765	37	1.0000	1.0000	0.9730
3	765	37	1.0000	1.0000	0.9730
4	765	37	0.9730	0.9730	0.9730
5	768	36	1.0000	1.0000	0.9722
6	768	36	1.0000	1.0000	0.9722
7	768	36	1.0000	1.0000	0.9722
8	768	36	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	**23**	**30**	**0.767**

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,454
CV mean ± std	0.9932 ± 0.0179
Hold-out test acc / loss	1.0000 · 0.0095
Batch-test acc	0.7000 (21/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	765	37	1.0000	1.0000	0.9730
2	765	37	1.0000	1.0000	0.9730
3	765	37	1.0000	1.0000	1.0000
4	765	37	0.9459	0.9459	0.9189
5	768	36	1.0000	1.0000	0.9722
6	768	36	1.0000	1.0000	0.9444
7	768	36	1.0000	1.0000	1.0000
8	768	36	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	0	5	0.000
Drum_Roll	1	5	0.200
Overall	**21**	**30**	**0.700**

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	203,878
CV mean ± std	0.9931 ± 0.0119
Hold-out test acc / loss	1.0000 · 0.0005
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	765	37	1.0000	1.0000	1.0000
2	765	37	1.0000	1.0000	1.0000
3	765	37	1.0000	1.0000	1.0000
4	765	37	0.9730	0.9730	0.9459
5	768	36	0.9722	0.9722	0.9722
6	768	36	1.0000	1.0000	0.9722
7	768	36	1.0000	1.0000	0.9722
8	768	36	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	**23**	**30**	**0.767**

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,190
CV mean ± std	0.9966 ± 0.0089
Hold-out test acc / loss	1.0000 · 0.0001
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	765	37	1.0000	1.0000	1.0000
2	765	37	1.0000	1.0000	1.0000
3	765	37	1.0000	1.0000	1.0000
4	765	37	0.9730	0.9730	0.9730
5	768	36	1.0000	1.0000	1.0000
6	768	36	1.0000	1.0000	1.0000
7	768	36	1.0000	1.0000	1.0000
8	768	36	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	3	5	0.600
Drum_Roll	0	5	0.000
Overall	**23**	**30**	**0.767**

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,622
CV mean ± std	0.9966 ± 0.0089
Hold-out test acc / loss	1.0000 · 0.0008
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	765	37	1.0000	1.0000	0.9459
2	765	37	1.0000	1.0000	0.9730
3	765	37	1.0000	1.0000	1.0000
4	765	37	0.9730	0.9730	0.9730
5	768	36	1.0000	1.0000	0.9722
6	768	36	1.0000	1.0000	1.0000
7	768	36	1.0000	1.0000	0.8611
8	768	36	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	**23**	**30**	**0.767**

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,390
CV mean ± std	0.9865 ± 0.0191
Hold-out test acc / loss	1.0000 · 0.0091
Batch-test acc	0.7000 (21/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	765	37	0.9730	0.9730	0.8108
2	765	37	0.9459	0.9730	0.8378
3	765	37	1.0000	1.0000	1.0000
4	765	37	0.9730	0.9730	0.9730
5	768	36	1.0000	1.0000	0.9722
6	768	36	1.0000	1.0000	1.0000
7	768	36	1.0000	1.0000	1.0000
8	768	36	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	1	5	0.200
Drum_Roll	1	5	0.200
Overall	**21**	**30**	**0.700**

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,510
CV mean ± std	0.9560 ± 0.1063
Hold-out test acc / loss	1.0000 · 0.0011
Batch-test acc	0.6667 (20/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	765	37	1.0000	1.0000	0.9730
2	765	37	1.0000	1.0000	1.0000
3	765	37	1.0000	1.0000	0.9730
4	765	37	0.6757	0.7027	0.6486
5	768	36	1.0000	1.0000	0.9444
6	768	36	0.9722	0.9722	0.9444
7	768	36	1.0000	1.0000	0.9722
8	768	36	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*33*
weighted avg	*1.000*	*1.000*	*1.000*	*33*

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	0	5	0.000
Drum_Roll	1	5	0.200
Overall	**20**	**30**	**0.667**